



Qualification Pack

Assistant Electrician (Domestic cum Industrial)

QP Code: MSME/PSS/Q0201

Version: 1.0

NSQF Level: 3.0

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Qualification Pack

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MSME/PSS/Q0201: Assistant Electrician (Domestic cum Industrial)

Brief Job Description

Learners who attain this qualification are competent to carry out electrical installation, maintenance at domestic and industrial site. Learned candidates are qualified to work as Assistant Electrician. Qualified learner can also start their own electrical workshop or choose to be an entrepreneur.

Personal Attributes

electrical installation, maintenance at domestic and industrial site.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MSME/PSS/N0203: Acquire the concepts of Basic Electrical](#)
2. [MSME/PSS/N0206: Measure Electrical Parameters \(Voltage, Current, Power etc.\)](#)
3. [MSME/PSS/N0215: Repair and Maintenance of Electrical Machines](#)
4. [MSME/PSS/N0214: Power Equipment Testing and Maintenance](#)
5. [MSME/PSS/N0213: Power Equipment Testing and Maintenance](#)
6. [MSME/PSS/N0212: Perform Transformer Testing and Maintenance](#)
7. [MSME/PSS/N0211: Perform Transformer Testing and Maintenance](#)
8. [MSME/PSS/N0210: Prepare for Domestic and Industrial Electrical Wiring](#)
9. [MSME/PSS/N0209: Prepare for Domestic and Industrial Electrical Wiring](#)
10. [MSME/PSS/N0207: Measure Electrical Parameters \(Voltage, Current, Power etc.\)](#)
11. [MSME/PSS/N0216: Employability Skills](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Distribution



Qualification Pack

Occupation	ELECTRICAL REPAIR
Country	India
NSQF Level	3.0
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO 7411.0100
Minimum Educational Qualification & Experience	10th Class with NA of experience OR Previous relevant Qualification of NSQF Level 2 with 3 Years of experience OR Previous relevant Qualification of NSQF Level 2.5 with 1.5 years of experience
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	15 Years
Last Reviewed On	NA
Next Review Date	30/04/2027
NSQC Approval Date	30/04/2024
Version	1.0
Reference code on NQR	QG-03-PW-02422-2024-V1-MSME
NQR Version	1.0



Qualification Pack

MSME/PSS/N0203: Acquire the concepts of Basic Electrical

Description

After completion of course learner should be able to observe safety precautions while working with electricity. Describe advantage of electricity.

Scope

The scope covers the following :

- After completion of course learner should be able to observe safety precautions while working with electricity.

Elements and Performance Criteria

MSME/DIE/01 Acquire the concepts of Basic Electrical

To be competent, the user/individual on the job must be able to:

- PC1.** Observe safety precautions while working with electricity.
- PC2.** Describe advantage of electricity.
- PC3.** Explain Ohm's law.
- PC4.**
 - Explain DC series and DC parallel circuits and its utility in electrical technology.
- PC5.**
 - Measure power and consumed electrical energy in any electric load.
- PC6.** Describe basic principles of electro statics and electro dynamics.
- PC7.** Describe properties of the natural magnet and electro magnet.
- PC8.**
 - Define magnetic flux, leakage flux, magnetomotive force, magnetizing force and flux density.
- PC9.**
 - Define hysteresis losses and eddy current losses in magnetic materials.
- PC10.** Explain the working principles of cells and batteries.
- PC11.**
 - Identify the direction of mechanical force between current carrying conductor and magnetic field.
- PC12.**
 - Explain fundamental of AC, sinusoidal curve, alternating quantity, cycle, time period, frequency, instantaneous value, maximum value, average value, RMS value form factor.
- PC13.**
 - Explain behavior of AC with resistive load, inductive load and capacitor load.
- PC14.** Describe star and delta connection in three phase system.
- PC15.** Identify phase and neutral in three phase four wire system.
- PC16.**
 - Describe different types of electrical engineering materials and their uses.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/DIE/01Acquire the concepts of Basic Electrical</i>	100	-	-	-
PC1. Observe safety precautions while working with electricity.	-	-	-	-
PC2. Describe advantage of electricity.	-	-	-	-
PC3. Explain Ohm's law.	-	-	-	-
PC4. • Explain DC series and DC parallel circuits and its utility in • electrical technology.	-	-	-	-
PC5. • Measure power and consumed electrical energy in any electric • load.	-	-	-	-
PC6. Describe basic principles of electro statics and electro dynamics.	-	-	-	-
PC7. Describe properties of the natural magnet and electro magnet.	-	-	-	-
PC8. • Define magnetic flux, leakage flux, magnetomotive force, • magnetizing force and flux density.	-	-	-	-
PC9. • Define hysteresis losses and eddy current losses in magnetic • materials.	-	-	-	-
PC10. Explain the working principles of cells and batteries.	-	-	-	-
PC11. • Identify the direction of mechanical force between current • carrying conductor and magnetic field.	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. • Explain fundamental of AC, sinusoidal curve, alternating • quantity, cycle, time period, frequency, instantaneous value, • maximum value, average value, RMS value form factor.	-	-	-	-
PC13. • Explain behavior of AC with resistive load, inductive load and • capacitor load.	-	-	-	-
PC14. Describe star and delta connection in three phase system.	-	-	-	-
PC15. Identify phase and neutral in three phase four wire system.	-	-	-	-
PC16. • Describe different types of electrical engineering materials and • their uses.	-	-	-	-
NOS Total	100	-	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/PSS/N0203
NOS Name	Acquire the concepts of Basic Electrical
Sector	Power
Sub-Sector	
Occupation	ELECTRICAL REPAIR
NSQF Level	3.0
Credits	2
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/PSS/N0206: Measure Electrical Parameters (Voltage, Current, Power etc.)

Description

After completion of course learner should be able to Explain safety precautions in handling tools and equipment.

Scope

The scope covers the following :

- After completion of course learner should be able to Explain safety precautions in handling tools and equipment.

Elements and Performance Criteria

MSME/DIE/02 Measure Electrical Parameters (Voltage, Current, Power etc.)

To be competent, the user/individual on the job must be able to:

- PC1.** Explain safety precautions in handling tools and equipment.
- PC2.** Identify the various tools used by an electrician.
- PC3.** Select the proper tool.
- PC4.** Use and maintain the tools used by an electrician.
- PC5.** Give the specification of tools for different applications.
- PC6.**
 - Measure the length and diameter with Vernier Calipers and
 - Micrometer.
- PC7.** Select and use AC/DC voltmeter, Ammeter as per requirements.
- PC8.** State the working principle of CT & PT and describe their use.
- PC9.** Measure power using wattmeter.
- PC10.**
 - Describe the use of DC regulated power supply, auto
 - transformer/variacs.
- PC11.**
 - Use electronic measuring instruments such as multimeter and
 - CRO etc. for electrical measurements.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/DIE/02 Measure Electrical Parameters (Voltage, Current, Power etc.)</i>	100	-	-	-
PC1. Explain safety precautions in handling tools and equipment.	-	-	-	-
PC2. Identify the various tools used by an electrician.	-	-	-	-
PC3. Select the proper tool.	-	-	-	-
PC4. Use and maintain the tools used by an electrician.	-	-	-	-
PC5. Give the specification of tools for different applications.	-	-	-	-
PC6. • Measure the length and diameter with Vernier Calipers and • Micrometer.	-	-	-	-
PC7. Select and use AC/DC voltmeter, Ammeter as per requirements.	-	-	-	-
PC8. State the working principle of CT & PT and describe their use.	-	-	-	-
PC9. Measure power using wattmeter.	-	-	-	-
PC10. • Describe the use of DC regulated power supply, auto • transformer/variatics.	-	-	-	-
PC11. • Use electronic measuring instruments such as multimeter and • CRO etc. for electrical measurements.	-	-	-	-
NOS Total	100	-	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/PSS/N0206
NOS Name	Measure Electrical Parameters (Voltage, Current, Power etc.)
Sector	Power
Sub-Sector	
Occupation	ELECTRICAL REPAIR
NSQF Level	3.0
Credits	1
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/PSS/N0215: Repair and Maintenance of Electrical Machines

Description

After completion of course Learner should be able to Explain the construction and operating principle of DC generator.

Scope

The scope covers the following :

- After completion of course Learner should be able to Explain the construction and operating principle of DC generator.

Elements and Performance Criteria

MSME/DIE/06 Repair and Maintenance of Electrical Machines

To be competent, the user/individual on the job must be able to:

- PC1.** • Explain the construction and operating principle of DC generator.
- PC2.** Explain the construction and operating principle of DC motor.
- PC3.** Carry out, servicing and repair of DC generators.
- PC4.** Carry out, servicing and repair of DC motors.
- PC5.** Explain construction, operating principle of transformer.
- PC6.** Explain construction and operating principle of alternator.
- PC7.** Repair faults in alternator and install the alternator.
- PC8.** • Explain construction and operating principle of three phase induction motor.
- PC9.** Test and identify the terminal of a DC generator.
- PC10.** Test and identify the terminal of a DC motor.
- PC11.** Test and identify the terminals of three phase induction motor.
- PC12.** • Test a three-phase induction motor for insulation resistance and earth effectiveness.
- PC13.** Connect with suitable starter, to start and run the DC motor.
- PC14.** • Connect with suitable starter to start and run three phase induction motor.
- PC15.** • Explain the construction, operating principle of different single phase or fractional horsepower (HP) motors.
- PC16.** • Identify the parts for repairing and servicing the fractional HP motor.
- PC17.** Troubleshoot different type of starter for DC motor.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/DIE/06 Repair and Maintenance of Electrical Machines</i>	100	-	-	-
PC1. • Explain the construction and operating principle of DC generator.	-	-	-	-
PC2. Explain the construction and operating principle of DC motor.	-	-	-	-
PC3. Carry out, servicing and repair of DC generators.	-	-	-	-
PC4. Carry out, servicing and repair of DC motors.	-	-	-	-
PC5. Explain construction, operating principle of transformer.	-	-	-	-
PC6. Explain construction and operating principle of alternator.	-	-	-	-
PC7. Repair faults in alternator and install the alternator.	-	-	-	-
PC8. • Explain construction and operating principle of three phase induction motor.	-	-	-	-
PC9. Test and identify the terminal of a DC generator.	-	-	-	-
PC10. Test and identify the terminal of a DC motor.	-	-	-	-
PC11. Test and identify the terminals of three phase induction motor.	-	-	-	-
PC12. • Test a three-phase induction motor for insulation resistance and • earth effectiveness.	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Connect with suitable starter, to start and run the DC motor.	-	-	-	-
PC14. - Connect with suitable starter to start and run three phase - induction motor.	-	-	-	-
PC15. - Explain the construction, operating principle of different single - phase or fractional horsepower (HP) motors.	-	-	-	-
PC16. - Identify the parts for repairing and servicing the fractional HP - motor.	-	-	-	-
PC17. Troubleshoot different type of starter for DC motor.	-	-	-	-
NOS Total	100	-	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/PSS/N0215
NOS Name	Repair and Maintenance of Electrical Machines
Sector	Power
Sub-Sector	
Occupation	ELECTRICAL REPAIR
NSQF Level	3.0
Credits	2
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



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MSME/PSS/N0214: Power Equipment Testing and Maintenance

Description

After completion of course Learner should be able to Design, manufacture of multi tapping auto transformer.

Scope

The scope covers the following :

- After completion of course Learner should be able to Design, manufacture of multi tapping auto transformer.

Elements and Performance Criteria

MSME/DIE/05 Power Equipment Testing and Maintenance

To be competent, the user/individual on the job must be able to:

- PC1.** Design, manufacture of multi tapping auto transformer.
- PC2.**
 - Explain construction and working principle of auto cut voltage stabilizer.
- PC3.** Assembly of auto cut voltage stabilizer circuit using DPDT switch.
- PC4.** Fault finding and repairing of auto cut voltage stabilizer.
- PC5.** Auto transformer designing for automatic voltage stabilizer.
- PC6.**
 - Working principle of automatic voltage stabilizer using DC supply.
- PC7.** Setting or alignment of automatic voltage stabilizer with variacs.
- PC8.** Fault finding and repairing of automatic voltage stabilizer.
- PC9.**
 - Explain different types of power rectification circuit with silicon diodes.
- PC10.**
 - Explain different types of power rectification circuit with different types of filter circuits.
- PC11.**
 - Explain definitions of cell and battery, primary battery, secondary battery.
- PC12.** Reconditioning of lead acid battery.
- PC13.** Design and assemble different types of battery charger.
- PC14.** Fault finding and repairing of different types of battery charger.
- PC15.** Explain circuit diagram of an UPS.
- PC16.** Repair and maintenance of UPS.
- PC17.** Explain of basic working principle of inverter.
- PC18.**
 - Explain different type of 500W MOSFET based inverter circuit with charger.
- PC19.** Assemble different types of inverter circuit with charger.
- PC20.** Install different types of inverter circuit with charger.



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PC21. Fault finding of different types of inverter circuit with charger.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/DIE/05 Power Equipment Testing and Maintenance</i>	-	100	-	-
PC1. Design, manufacture of multi tapping auto transformer.	-	-	-	-
PC2. • Explain construction and working principle of auto cut voltage • stabilizer.	-	-	-	-
PC3. Assembly of auto cut voltage stabilizer circuit using DPDT switch.	-	-	-	-
PC4. Fault finding and repairing of auto cut voltage stabilizer.	-	-	-	-
PC5. Auto transformer designing for automatic voltage stabilizer.	-	-	-	-
PC6. • Working principle of automatic voltage stabilizer using DC • supply.	-	-	-	-
PC7. Setting or alignment of automatic voltage stabilizer with variacs.	-	-	-	-
PC8. Fault finding and repairing of automatic voltage stabilizer.	-	-	-	-
PC9. • Explain different types of power rectification circuit with silicon • diodes.	-	-	-	-
PC10. • Explain different types of power rectification circuit with • different types of filter circuits.	-	-	-	-
PC11. • Explain definitions of cell and battery, primary battery, • secondary battery.	-	-	-	-
PC12. Reconditioning of lead acid battery.	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Design and assemble different types of battery charger.	-	-	-	-
PC14. Fault finding and repairing of different types of battery charger.	-	-	-	-
PC15. Explain circuit diagram of an UPS.	-	-	-	-
PC16. Repair and maintenance of UPS.	-	-	-	-
PC17. Explain of basic working principle of inverter.	-	-	-	-
PC18. - Explain different type of 500W MOSFET based inverter circuit - with charger.	-	-	-	-
PC19. Assemble different types of inverter circuit with charger.	-	-	-	-
PC20. Install different types of inverter circuit with charger.	-	-	-	-
PC21. Fault finding of different types of inverter circuit with charger.	-	-	-	-
NOS Total	-	100	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/PSS/N0214
NOS Name	Power Equipment Testing and Maintenance
Sector	Power
Sub-Sector	
Occupation	ELECTRICAL REPAIR
NSQF Level	3.0
Credits	2
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



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MSME/PSS/N0213: Power Equipment Testing and Maintenance

Description

After completion of course Learner should be able to Design, manufacture of multi tapping auto transformer.

Scope

The scope covers the following :

- After completion of course Learner should be able to Design, manufacture of multi tapping auto transformer.

Elements and Performance Criteria

MSME/DIE/05 Power Equipment Testing and Maintenance

To be competent, the user/individual on the job must be able to:

- PC1.** Design, manufacture of multi tapping auto transformer.
- PC2.**
 - Explain construction and working principle of auto cut voltage stabilizer.
- PC3.** Assembly of auto cut voltage stabilizer circuit using DPDT switch.
- PC4.** Fault finding and repairing of auto cut voltage stabilizer.
- PC5.** Auto transformer designing for automatic voltage stabilizer.
- PC6.**
 - Working principle of automatic voltage stabilizer using DC supply.
- PC7.** Setting or alignment of automatic voltage stabilizer with variacs.
- PC8.** Fault finding and repairing of automatic voltage stabilizer.
- PC9.**
 - Explain different types of power rectification circuit with silicon diodes.
- PC10.**
 - Explain different types of power rectification circuit with different types of filter circuits.
- PC11.**
 - Explain definitions of cell and battery, primary battery, secondary battery.
- PC12.** Reconditioning of lead acid battery.
- PC13.** Design and assemble different types of battery charger.
- PC14.** Fault finding and repairing of different types of battery charger.
- PC15.** Explain circuit diagram of an UPS.
- PC16.** Repair and maintenance of UPS.
- PC17.** Explain of basic working principle of inverter.
- PC18.**
 - Explain different type of 500W MOSFET based inverter circuit with charger.
- PC19.** Assemble different types of inverter circuit with charger.
- PC20.** Install different types of inverter circuit with charger.



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PC21. Fault finding of different types of inverter circuit with charger.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/DIE/05 Power Equipment Testing and Maintenance</i>	100	-	-	-
PC1. Design, manufacture of multi tapping auto transformer.	-	-	-	-
PC2. • Explain construction and working principle of auto cut voltage • stabilizer.	-	-	-	-
PC3. Assembly of auto cut voltage stabilizer circuit using DPDT switch.	-	-	-	-
PC4. Fault finding and repairing of auto cut voltage stabilizer.	-	-	-	-
PC5. Auto transformer designing for automatic voltage stabilizer.	-	-	-	-
PC6. • Working principle of automatic voltage stabilizer using DC • supply.	-	-	-	-
PC7. Setting or alignment of automatic voltage stabilizer with variacs.	-	-	-	-
PC8. Fault finding and repairing of automatic voltage stabilizer.	-	-	-	-
PC9. • Explain different types of power rectification circuit with silicon • diodes.	-	-	-	-
PC10. • Explain different types of power rectification circuit with • different types of filter circuits.	-	-	-	-
PC11. • Explain definitions of cell and battery, primary battery, • secondary battery.	-	-	-	-
PC12. Reconditioning of lead acid battery.	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Design and assemble different types of battery charger.	-	-	-	-
PC14. Fault finding and repairing of different types of battery charger.	-	-	-	-
PC15. Explain circuit diagram of an UPS.	-	-	-	-
PC16. Repair and maintenance of UPS.	-	-	-	-
PC17. Explain of basic working principle of inverter.	-	-	-	-
PC18. • Explain different type of 500W MOSFET based inverter circuit • with charger.	-	-	-	-
PC19. Assemble different types of inverter circuit with charger.	-	-	-	-
PC20. Install different types of inverter circuit with charger.	-	-	-	-
PC21. Fault finding of different types of inverter circuit with charger.	-	-	-	-
NOS Total	100	-	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/PSS/N0213
NOS Name	Power Equipment Testing and Maintenance
Sector	Power
Sub-Sector	
Occupation	ELECTRICAL REPAIR
NSQF Level	3.0
Credits	1
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/PSS/N0212: Perform Transformer Testing and Maintenance

Description

After completion of course Learner should be able to Draw the symbols, abbreviations used in transformer designing.

Scope

The scope covers the following :

- After completion of course Learner should be able to Draw the symbols, abbreviations used in transformer designing.

Elements and Performance Criteria

MSME/DIE/04 Perform Transformer Testing and Maintenance

To be competent, the user/individual on the job must be able to:

- PC1.** Draw the symbols, abbreviations used in transformer designing.
- PC2.**
 - Define magnetic flux, leakage flux, magnetizing force and flux density.
- PC3.** Explain working principle of a simple transformer.
- PC4.**
 - Define the hysteresis losses and eddy current losses in magnetic material.
- PC5.** Describe the properties of core material and their types.
- PC6.** Select the proper core material size & shape.
- PC7.** Explain basic considerations of winding.
- PC8.** Explain characteristics of super enamel copper wire.
- PC9.** Select the proper size of copper conductor for a transformer.
- PC10.** Select the proper insulation for winding a coil.
- PC11.**
 - Wind a coil for required transformer and assemble core along with coil.
- PC12.** Explain different tests required for a transformer.
- PC13.**
 - Explain temperature rise and its effect on the performance of transformer.
- PC14.** State the design equations of the transformer.
- PC15.** Design the main transformer up to 5KVA.
- PC16.** Design the transformer for voltage stabilizer.
- PC17.** Operate, repair and maintain the winding machine.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/DIE/04 Perform Transformer Testing and Maintenance</i>	-	100	-	-
PC1. Draw the symbols, abbreviations used in transformer designing.	-	-	-	-
PC2. - Define magnetic flux, leakage flux, magnetizing force and flux - density.	-	-	-	-
PC3. Explain working principle of a simple transformer.	-	-	-	-
PC4. - Define the hysteresis losses and eddy current losses in magnetic - material.	-	-	-	-
PC5. Describe the properties of core material and their types.	-	-	-	-
PC6. Select the proper core material size & shape.	-	-	-	-
PC7. Explain basic considerations of winding.	-	-	-	-
PC8. Explain characteristics of super enamel copper wire.	-	-	-	-
PC9. Select the proper size of copper conductor for a transformer.	-	-	-	-
PC10. Select the proper insulation for winding a coil.	-	-	-	-
PC11. - Wind a coil for required transformer and assemble core along - with coil.	-	-	-	-
PC12. Explain different tests required for a transformer.	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. • Explain temperature rise and its effect on the performance of • transformer.	-	-	-	-
PC14. State the design equations of the transformer.	-	-	-	-
PC15. Design the main transformer up to 5KVA.	-	-	-	-
PC16. Design the transformer for voltage stabilizer.	-	-	-	-
PC17. Operate, repair and maintain the winding machine.	-	-	-	-
NOS Total	-	100	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSME/PSS/N0212
NOS Name	Perform Transformer Testing and Maintenance
Sector	Power
Sub-Sector	
Occupation	ELECTRICAL REPAIR
NSQF Level	3.0
Credits	2
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/PSS/N0211: Perform Transformer Testing and Maintenance

Description

After completion of course Learner should be able to Draw the symbols, abbreviations used in transformer designing.

Scope

The scope covers the following :

- After completion of course Learner should be able to Draw the symbols, abbreviations used in transformer designing.

Elements and Performance Criteria

MSME/DIE/04 Perform Transformer Testing and Maintenance

To be competent, the user/individual on the job must be able to:

- PC1.** Draw the symbols, abbreviations used in transformer designing.
- PC2.**
 - Define magnetic flux, leakage flux, magnetizing force and flux density.
- PC3.** Explain working principle of a simple transformer.
- PC4.**
 - Define the hysteresis losses and eddy current losses in magnetic material.
- PC5.** Describe the properties of core material and their types.
- PC6.** Select the proper core material size & shape.
- PC7.** Explain basic considerations of winding.
- PC8.** Explain characteristics of super enamel copper wire.
- PC9.** Select the proper size of copper conductor for a transformer.
- PC10.** Select the proper insulation for winding a coil.
- PC11.**
 - Wind a coil for required transformer and assemble core along with coil.
- PC12.** Explain different tests required for a transformer.
- PC13.**
 - Explain temperature rise and its effect on the performance of transformer.
- PC14.** State the design equations of the transformer.
- PC15.** Design the main transformer up to 5KVA.
- PC16.** Design the transformer for voltage stabilizer.
- PC17.** Operate, repair and maintain the winding machine.



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/DIE/04 Perform Transformer Testing and Maintenance</i>	100	-	-	-
PC1. Draw the symbols, abbreviations used in transformer designing.	-	-	-	-
PC2. • Define magnetic flux, leakage flux, magnetizing force and flux • density.	-	-	-	-
PC3. Explain working principle of a simple transformer.	-	-	-	-
PC4. • Define the hysteresis losses and eddy current losses in magnetic • material.	-	-	-	-
PC5. Describe the properties of core material and their types.	-	-	-	-
PC6. Select the proper core material size & shape.	-	-	-	-
PC7. Explain basic considerations of winding.	-	-	-	-
PC8. Explain characteristics of super enamel copper wire.	-	-	-	-
PC9. Select the proper size of copper conductor for a transformer.	-	-	-	-
PC10. Select the proper insulation for winding a coil.	-	-	-	-
PC11. • Wind a coil for required transformer and assemble core along • with coil.	-	-	-	-
PC12. Explain different tests required for a transformer.	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. • Explain temperature rise and its effect on the performance of • transformer.	-	-	-	-
PC14. State the design equations of the transformer.	-	-	-	-
PC15. Design the main transformer up to 5KVA.	-	-	-	-
PC16. Design the transformer for voltage stabilizer.	-	-	-	-
PC17. Operate, repair and maintain the winding machine.	-	-	-	-
NOS Total	100	-	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MSME/PSS/N0211
NOS Name	Perform Transformer Testing and Maintenance
Sector	Power
Sub-Sector	
Occupation	ELECTRICAL REPAIR
NSQF Level	3.0
Credits	1
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/PSS/N0210: Prepare for Domestic and Industrial Electrical Wiring

Description

After completion of course learner should be able to Identify different types of wire joints and prepare different wire joints.

Scope

The scope covers the following :

- After completion of course learner should be able to Identify different types of wire joints and prepare different wire joints.

Elements and Performance Criteria

MSME/DIE/03 Prepare for Domestic and Industrial Electrical Wiring

To be competent, the user/individual on the job must be able to:

- PC1.** • Identify different types of wire joints and prepare different wire joints.
- PC2.** • Describe the specifications of different components and materials used in house wiring installation.
- PC3.** Explain the different type of wiring system.
- PC4.** Draw the different types of electrical wiring circuits.
- PC5.** • Choose proper type of wiring system as per customer's requirements.
- PC6.** Perform power wiring for industrial systems.
- PC7.** Design and wire up simple electrical circuit in the house.
- PC8.** Estimate the wiring installation in a house.
- PC9.** • Calculate proper size of wire as per load connected in the wiring installation.
- PC10.** • Explain the different types of fuses and calculate the size of fuse wire.
- PC11.** • Explain the electrical earthing system and its need for wiring installation.
- PC12.** Draw a wiring diagram for a house.
- PC13.** Test the wiring installation with megger.
- PC14.** Identify the fault in wiring installation and repair the fault.
- PC15.** Identify the different types of power wiring accessories.
- PC16.** Describe the different types of substations.
- PC17.** Explain the provisions of Indian Electricity Rules 1956.
- PC18.** Perform electrical shock treatment.
- PC19.** Understand the Calculation of wire selection.
- PC20.** Understand types component used according to load.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/DIE/03 Prepare for Domestic and Industrial Electrical Wiring</i>	100	-	-	-
PC1. • Identify different types of wire joints and prepare different wire joints.	-	-	-	-
PC2. • Describe the specifications of different components and materials used in house wiring installation.	-	-	-	-
PC3. Explain the different type of wiring system.	-	-	-	-
PC4. Draw the different types of electrical wiring circuits.	-	-	-	-
PC5. • Choose proper type of wiring system as per customer's requirements.	-	-	-	-
PC6. Perform power wiring for industrial systems.	-	-	-	-
PC7. Design and wire up simple electrical circuit in the house.	-	-	-	-
PC8. Estimate the wiring installation in a house.	-	-	-	-
PC9. • Calculate proper size of wire as per load connected in the wiring • installation.	-	-	-	-
PC10. • Explain the different types of fuses and calculate the size of fuse • wire.	-	-	-	-
PC11. • Explain the electrical earthing system and its need for wiring • installation.	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Draw a wiring diagram for a house.	-	-	-	-
PC13. Test the wiring installation with megger.	-	-	-	-
PC14. Identify the fault in wiring installation and repair the fault.	-	-	-	-
PC15. Identify the different types of power wiring accessories.	-	-	-	-
PC16. Describe the different types of substations.	-	-	-	-
PC17. Explain the provisions of Indian Electricity Rules 1956.	-	-	-	-
PC18. Perform electrical shock treatment.	-	-	-	-
PC19. Understand the Calculation of wire selection.	-	-	-	-
PC20. Understand types component used according to load.	-	-	-	-
NOS Total	100	-	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MSME/PSS/N0210
NOS Name	Prepare for Domestic and Industrial Electrical Wiring
Sector	Power
Sub-Sector	
Occupation	ELECTRICAL REPAIR
NSQF Level	3.0
Credits	1
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/PSS/N0209: Prepare for Domestic and Industrial Electrical Wiring

Description

After completion of course learner should be able to Identify different types of wire joints and prepare different wire joints.

Scope

The scope covers the following :

- After completion of course learner should be able to Identify different types of wire joints and prepare different wire joints.

Elements and Performance Criteria

MSME/DIE/03 Prepare for Domestic and Industrial Electrical Wiring

To be competent, the user/individual on the job must be able to:

- PC1.** • Identify different types of wire joints and prepare different wire joints.
- PC2.** • Describe the specifications of different components and materials used in house wiring installation.
- PC3.** Explain the different type of wiring system.
- PC4.** Draw the different types of electrical wiring circuits.
- PC5.** • Choose proper type of wiring system as per customer's requirements.
- PC6.** Perform power wiring for industrial systems.
- PC7.** Design and wire up simple electrical circuit in the house.
- PC8.** Estimate the wiring installation in a house.
- PC9.** • Calculate proper size of wire as per load connected in the wiring installation.
- PC10.** • Explain the different types of fuses and calculate the size of fuse wire.
- PC11.** • Explain the electrical earthing system and its need for wiring installation.
- PC12.** Draw a wiring diagram for a house.
- PC13.** Test the wiring installation with megger.
- PC14.** Identify the fault in wiring installation and repair the fault.
- PC15.** Identify the different types of power wiring accessories.
- PC16.** Describe the different types of substations.
- PC17.** Explain the provisions of Indian Electricity Rules 1956.
- PC18.** Perform electrical shock treatment.
- PC19.** Understand the Calculation of wire selection.
- PC20.** Understand types component used according to load.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/DIE/03 Prepare for Domestic and Industrial Electrical Wiring</i>	-	100	-	-
PC1. Identify different types of wire joints and prepare different wire joints.	-	-	-	-
PC2. Describe the specifications of different components and materials used in house wiring installation.	-	-	-	-
PC3. Explain the different type of wiring system.	-	-	-	-
PC4. Draw the different types of electrical wiring circuits.	-	-	-	-
PC5. Choose proper type of wiring system as per customer's requirements.	-	-	-	-
PC6. Perform power wiring for industrial systems.	-	-	-	-
PC7. Design and wire up simple electrical circuit in the house.	-	-	-	-
PC8. Estimate the wiring installation in a house.	-	-	-	-
PC9. Calculate proper size of wire as per load connected in the wiring installation.	-	-	-	-
PC10. Explain the different types of fuses and calculate the size of fuse wire.	-	-	-	-
PC11. Explain the electrical earthing system and its need for wiring installation.	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Draw a wiring diagram for a house.	-	-	-	-
PC13. Test the wiring installation with megger.	-	-	-	-
PC14. Identify the fault in wiring installation and repair the fault.	-	-	-	-
PC15. Identify the different types of power wiring accessories.	-	-	-	-
PC16. Describe the different types of substations.	-	-	-	-
PC17. Explain the provisions of Indian Electricity Rules 1956.	-	-	-	-
PC18. Perform electrical shock treatment.	-	-	-	-
PC19. Understand the Calculation of wire selection.	-	-	-	-
PC20. Understand types component used according to load.	-	-	-	-
NOS Total	-	100	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MSME/PSS/N0209
NOS Name	Prepare for Domestic and Industrial Electrical Wiring
Sector	Power
Sub-Sector	
Occupation	ELECTRICAL REPAIR
NSQF Level	3.0
Credits	5
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/PSS/N0207: Measure Electrical Parameters (Voltage, Current, Power etc.)

Description

After completion of course learner should be able to Explain safety precautions in handling tools and equipment.

Scope

The scope covers the following :

- After completion of course learner should be able to Explain safety precautions in handling tools and equipment.

Elements and Performance Criteria

MSME/DIE/02 Measure Electrical Parameters (Voltage, Current, Power etc.)

To be competent, the user/individual on the job must be able to:

- PC1.** Explain safety precautions in handling tools and equipment.
- PC2.** Identify the various tools used by an electrician.
- PC3.** Select the proper tool.
- PC4.** Use and maintain the tools used by an electrician.
- PC5.** Give the specification of tools for different applications.
- PC6.**
 - Measure the length and diameter with Vernier Calipers and
 - Micrometer.
- PC7.** Select and use AC/DC voltmeter, Ammeter as per requirements.
- PC8.** State the working principle of CT & PT and describe their use.
- PC9.** Measure power using wattmeter.
- PC10.**
 - Describe the use of DC regulated power supply, auto
 - transformer/variacs.
- PC11.**
 - Use electronic measuring instruments such as multimeter and
 - CRO etc. for electrical measurements.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/DIE/02 Measure Electrical Parameters (Voltage, Current, Power etc.)</i>	-	100	-	-
PC1. Explain safety precautions in handling tools and equipment.	-	-	-	-
PC2. Identify the various tools used by an electrician.	-	-	-	-
PC3. Select the proper tool.	-	-	-	-
PC4. Use and maintain the tools used by an electrician.	-	-	-	-
PC5. Give the specification of tools for different applications.	-	-	-	-
PC6. • Measure the length and diameter with Vernier Calipers and • Micrometer.	-	-	-	-
PC7. Select and use AC/DC voltmeter, Ammeter as per requirements.	-	-	-	-
PC8. State the working principle of CT & PT and describe their use.	-	-	-	-
PC9. Measure power using wattmeter.	-	-	-	-
PC10. • Describe the use of DC regulated power supply, auto • transformer/variatics.	-	-	-	-
PC11. • Use electronic measuring instruments such as multimeter and • CRO etc. for electrical measurements.	-	-	-	-
NOS Total	-	100	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MSME/PSS/N0207
NOS Name	Measure ElectricalParameters (Voltage,Current, Power etc.)
Sector	Power
Sub-Sector	
Occupation	ELECTRICAL REPAIR
NSQF Level	3.0
Credits	2
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

MSME/PSS/N0216: Employability Skills

Description

After completing this programme participants will be able to Discuss the importance of Employability Skills in meeting the job requirements

Scope

The scope covers the following :

- After completing this programme participants will be able to Discuss the importance of Employability Skills in meeting the job requirements

Elements and Performance Criteria

MSME/ES/01 Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** • Discuss the importance of Employability Skills in meeting the job requirements.
- PC2.** • Explain constitutional values, civic rights, duties, citizenship, responsibility towards society etc. that are required to be followed to become a responsible citizen.
- PC3.** • Show how to practice different environmentally sustainable practices.
- PC4.** Discuss 21st century skills.
- PC5.** • Display positive attitude, self -motivation, problem solving, time management skills and continuous learning mindset in different situations.
- PC6.** • Use appropriate basic English sentences/phrases while speaking.
- PC7.** • Demonstrate how to communicate in a well -mannered way with others.
- PC8.** Demonstrate working with others in a team.
- PC9.** • Show how to conduct oneself appropriately with all genders and PwD
- PC10.** • Discuss the significance of reporting sexual harassment issues in time
- PC11.** • Discuss the significance of using financial products and services safely and securely.
- PC12.** • Explain the importance of managing expenses, income, and savings.
- PC13.** • Explain the significance of approaching the concerned authorities in time for any exploitation as per legal rights and laws.
- PC14.** • Show how to operate digital devices and use the associated applications and features, safely and securely.



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- PC15.** • Discuss the significance of using internet for browsing, accessing
• social media platforms, safely and Securely.
- PC16.** • Discuss the need for identifying opportunities for potential
• business, sources for arranging money and potential legal and
• financial challenges.
- PC17.** Differentiate between types of customers.
- PC18.** • Explain the significance of identifying customer needs and
• addressing them.
- PC19.** • Discuss the significance of maintaining hygiene and dressing
• appropriately.
- PC20.** Create a biodata.
- PC21.** Use various sources to search and apply for jobs.
- PC22.** • Discuss the significance of dressing up neatly and maintaining
• hygiene for an interview.
- PC23.** • Discuss how to search and register for apprenticeship
• opportunities.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>MSME/ES/01 Employability Skills</i>	100	-	-	-
PC1. Discuss the importance of Employability Skills in meeting the job requirements.	-	-	-	-
PC2. - Explain constitutional values, civic rights, duties, citizenship, - responsibility towards society etc. that are required to be - followed to become a responsible citizen.	-	-	-	-
PC3. - Show how to practice different environmentally sustainable - practices.	-	-	-	-
PC4. Discuss 21st century skills.	-	-	-	-
PC5. - Display positive attitude, self -motivation, problem solving, time - management skills and continuous learning mindset in different - situations.	-	-	-	-
PC6. - Use appropriate basic English sentences/phrases while - speaking.	-	-	-	-
PC7. - Demonstrate how to communicate in a well -mannered way - with others.	-	-	-	-
PC8. Demonstrate working with others in a team.	-	-	-	-
PC9. - Show how to conduct oneself appropriately with all genders and - PwD	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. - Discuss the significance of reporting sexual harassment issues in - time	-	-	-	-
PC11. - Discuss the significance of using financial products and services - safely and securely.	-	-	-	-
PC12. - Explain the importance of managing expenses, income, and - savings.	-	-	-	-
PC13. - Explain the significance of approaching the concerned - authorities in time for any exploitation as per legal rights and - laws.	-	-	-	-
PC14. - Show how to operate digital devices and use the associated - applications and features, safely and securely.	-	-	-	-
PC15. - Discuss the significance of using internet for browsing, accessing - social media platforms, safely and Securely.	-	-	-	-
PC16. - Discuss the need for identifying opportunities for potential - business, sources for arranging money and potential legal and - financial challenges.	-	-	-	-
PC17. Differentiate between types of customers.	-	-	-	-
PC18. - Explain the significance of identifying customer needs and - addressing them.	-	-	-	-
PC19. - Discuss the significance of maintaining hygiene and dressing - appropriately.	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC20. Create a biodata.	-	-	-	-
PC21. Use various sources to search and apply for jobs.	-	-	-	-
PC22. - Discuss the significance of dressing up neatly and maintaining - hygiene for an interview.	-	-	-	-
PC23. - Discuss how to search and register for apprenticeship - opportunities.	-	-	-	-
NOS Total	100	-	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MSME/PSS/N0216
NOS Name	EmployabilitySkills
Sector	Power
Sub-Sector	
Occupation	ELECTRICAL REPAIR
NSQF Level	3.0
Credits	1
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

As per QP

Minimum Aggregate Passing % at QP Level : 40

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MSME/PSS/N0203.Acquire the concepts of Basic Electrical	100	0	0	0	100	10
MSME/PSS/N0206.Measure Electrical Parameters (Voltage,Current, Power etc.)	100	0	0	0	100	10
MSME/PSS/N0215.Repair and Maintenance of Electrical Machines	100	0	0	0	100	10
MSME/PSS/N0214.Power Equipment Testing and Maintenance	0	100	0	0	100	10
MSME/PSS/N0213.Power Equipment Testing and Maintenance	100	0	0	0	100	10
MSME/PSS/N0212.Perform Transformer Testing and Maintenance	0	100	0	0	100	10
MSME/PSS/N0211.Perform Transformer Testing and Maintenance	100	0	0	0	100	10
MSME/PSS/N0210.Prepare for Domestic and Industrial Electrical Wiring	100	0	0	0	100	10
MSME/PSS/N0209.Prepare for Domestic and Industrial Electrical Wiring	0	100	0	0	100	10
MSME/PSS/N0207.Measure ElectricalParameters (Voltage,Current, Power etc.)	0	100	0	0	100	10
MSME/PSS/N0216.EmployabilitySkills	100	0	0	0	100	0
Total	700	400	-	-	1100	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.